

Vision Empower Trust AI Skilling Program

MODULE 3

AI for VI

Vision Empower Trust



Agenda

Item	Details
Audience	Teachers, special educators, resource persons and school leaders working with visually impaired students
Duration	90 minutes
Mode	Hands on
Module purpose	This module will help teachers understand how AI can support persons with visual impairment in education, daily living, mobility, communication, and independent access to information. It will also help teachers understand the limitations of AI and the need for safe, responsible and verified use.
Session principle	This session is practical and teacher-centered, with a focus on using AI responsibly to create inclusive, accessible, and personalized learning resources. Teachers will learn through hands-on activities and refine prompts to achieve better AI-generated outcomes.

Learning Outcomes

- Explain how AI can support persons with visual impairment.
- Identify AI tools that help with reading, image description, navigation, translation, communication and learning.
- Understand how AI can act as a support for “substitution of vision” in some situations.
- Recognise the limitations and risks of using AI tools.
- Guide VI students to use AI safely and independently.
- Apply a simple verification habit before trusting AI output.

Pre-Assessment

- Can AI be used by the visually impaired?
- Do you know any of the tools that can be used by AI
- What are the guardrails and checks to be done when using AI as a VI person

Why AI for VI?

- AI can give access to visual information through speech, text, audio and other alternate user interfaces, making visual information accessible
- Special AI models are rigorously trained to perform many human functions that can be helpful for the VI community
 - Image recognition
 - Speech recognition
 - Speech to text, text to speech conversion
 - Language Translation, Transcription of text, audio or video
 - OCR by converting inaccessible text to digital text

AI in Assistive Technologies and AI as Assistive Technology

- Many Assistive Technology solutions are becoming AI enabled
 - Be My Eyes -> Be My AI
 - YouDescribe volunteer led Video Transcription -> AI led transcription with human verification
- AI by itself can perform functions that can help in accessibility
 - Seeing AI – Computer Vision to convert images to textual/ audio description
 - Generative AI to generate description of images as text / audio or creation of audio to text
 - Knowing about colour changes in chemistry experiments using AI tools
- AI can help in compensating vision loss to a certain extent

Some AI tools for VI users

- Generative AI Tools (Covered in Module 2)
 - Microsoft CoPilot
 - ChatGPT
 - Suno AI
 - Sarvam AI
- Special AI Tools
 - Be My AI
 - Microsoft Seeing AI
 - Google Lookout
 - Braille AI Tutors like <https://braille.academy/ai-tutor>
 - AI-Learners which help personalize learning for the students
- Some of these tools are described below.

Microsoft Seeing AI – A Visual Assistant for the Blind



- Assistance with daily tasks like reading text, describe photos or people, objects, scenes, currency, colours, products and images.
- It is designed specifically with and for blind and low-vision users
- Vision Empower Tutorials : https://www.youtube.com/watch?v=ba-AqvGH7ag&list=PLE5Q6U3y6x5YgzNeiae_hHvzml58tSiNR

Be My AI – AI Powered Assistant to Virtual Volunteer Be My Eyes



- Have you used Be My Eyes as a VI person or as a volunteer?
- Be My AI is a digital visual assistant built into the [Be My Eyes](#) app that uses artificial intelligence to describe images and answer questions for people who are blind or have low vision.
- **Take Pictures:** Capture photos using your phone camera to get detailed descriptions of your surroundings.
- **Ask Follow-Ups:** Chat with the AI to ask specific questions about objects, text, or details in the photo.
- **Share Images:** Send pictures directly from your photo gallery, social media, or messaging apps like WhatsApp using the share menu.
- **Human Backup: Connect with a live human volunteer if you need extra help or have urgent safety questions.**
- Vision Empower
Tutorials: <https://www.youtube.com/watch?v=5ifjPWXYZjs&list=PLE5Q6U3y6x5aDK964XeWeSEnInlEwlp56>

Microsoft CoPilot



- **Gen AI** that can **convert visual data into descriptive text**, reading screen content aloud, and supporting screen readers like NVDA and Narrator
- Users can paste **graphs, charts, or photos** into Copilot and ask it to describe visual elements in detail
- Screen Reading Support: Works with Windows Narrator and third-party screen readers to read web pages, emails, and documents aloud with natural voices
- Allows users to share their active screen or window and interact entirely via natural voice commands without needing to see the layout.
- Vision Empower Tutorials
: <https://www.youtube.com/watch?v=xZgO4QBKGrU&list=PLE5Q6U3y6x5boMKBiq0ZuvX-kDdp5nOn>

AI-enabled smart glasses

- Envision Glasses and Meta AI-enabled glasses are being explored for hands-free visual assistance
- Be My Eyes has also been integrated with Meta AI glasses, allowing users to receive visual assistance through wearable technology.



Risks and limitations

- **AI can be wrong** even when it sounds confident.
- Tools **can misread** text, miss details, describe an image incorrectly, or give incomplete information.
- They **should not be used as the final authority in high-risk situations**
- VI users should **avoid depending only on AI for medicines, dosage, allergens, road crossing, legal documents, exams, financial transactions, identity verification, or emergency decisions**
- Be My Eyes specifically says not to use Be My AI for scanning medicines, reading dosages, or other safety issues.
- Image descriptions may miss context, facial recognition may be incorrect
- Good lighting and contrast plays an important role in the proper description of images
- Privacy must be protected, personal, sensitive, confidential data should not be shared
- AI may increase overdependence, causing harm to creativity and critical thinking

Is it safe to rely on Be My AI for scanning medicine bottles?

No. Do not use Be My AI for scanning medicines, reading dosages, or other safety issues.

Should I use Be My AI with personal and financial information?

Please do not use Be My AI to process sensitive personal or financial information. This is one area where asking a trusted friend or family member for help is still the best course of action.

Are Copilot's responses always accurate?

Copilot aims to respond with reliable sources, but AI can make mistakes, and third-party content on the internet may not always be accurate or reliable. Copilot may misrepresent the information it finds, and you may see responses that sound convincing but are incomplete, inaccurate, or inappropriate. Use your judgment and double check facts before making decisions or taking action based on Copilot's responses. Reviewing Copilot's citations is a good place to start checking for accuracy.

Microsoft built a tool called Seeing AI. It used computer vision to help visually impaired people understand who was in a room. Age, emotion, gender, all of it.

Then law enforcement started downloading the same cognitive service.

They used it to identify people. The dataset wasn't built for that. The accuracy wasn't there. People got misidentified.

How do we use AI if it has pros and cons?

- Guardrails are built by applications to ensure accuracy for users, like Copilot
- Humans are accountable for what we use AI for and how we use its outputs
- Responsible, Informed and Conscious (RIC) Use of AI needs to be practiced
- Responsible :
 - Use AI in a manner that does not result in harm for yourself or others
- Informed :
 - Know that AI can make mistakes, can have bias, can make mistakes, does not have emotions
 - Know that AI can generate content without it being reality (hallucination)
- Conscious :
 - Use AI with full understanding of the accountability for actions based on the information or suggestions that are got from the AI solutions
- An **Responsible, Informed and Conscious** human in the loop is the best way to use AI
- Pause – Check – Verify – Decide

Let us all remember

- AI is not a substitute for
 - Professional advice from teachers, doctors, CA, lawyers etc.
 - Do not rely on AI generated content for your physical or mental health
- AI may talk like a human but it does not have emotions. The responses are generated to make the humans “feel like they are talking to another human”
- We should not develop overdependence on AI
- AI is not always accurate
 - Self judgement for accuracy is a must
 - Lag between current updates and the training data
 - Not being aware of the training the model has received
- Responsibility of using AI properly is on the humans
- When there is a need to get live data input, official sources information, access the sources directly without AI
- Use AI only when needed as there is an environmental cost for running AI models as well due to the large computing and data storage needs

Post-Assessment

- Can AI be used by the visually impaired?
- Do you know any of the tools that can be used by AI
- What are the guardrails and checks to be done when using AI as a VIpersion

AI

**can be a powerful assistant for the
humans when used in an
Responsible, Informed, and
Conscious
manner**

Thank You !

